

Lirui (Leroy) Wang

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Education:

University of Washington, Seattle, WA (GPA:**3.96**/Annual **Dean's list**)

- Bachelor of Science, Electrical Engineering, College of Engineering Expected Jun 2020
- Bachelor of Science, Computer Science, College of Arts and Sciences Expected Jun 2020

Work Experience:

Intern, DJI Technology, RoboMasters Department, Shenzhen, China

Jul 2017 – Aug 2017

- Worked on a robot and a drone (with Nvidia TX1 onboard) that would compete in a fully automatic robotic battle
- Programmed take-off platform, barrier tape and tag detection for the drone to retrieve info such as relative location
- Applied probabilistic algorithm to navigate and locate, and built a top-level state machine to strategize our robot

Intern, SZ OPEN Robotics Technology, Shenzhen, China

Jan 2016 – Mar 2016

- Designed the mechanics and electronics of a dancing robot and implemented the field tests with PID controllers
- Developed competition demo robots that can do black line tracking, color detection and route planning for undergraduates

Projects:

Robotic Alignment Project – UW Robotics and States Estimation Lab

June 2018 – Present

- Attempt to estimate the SE3 state of robot arm in real image by training the network on a synthetic dataset.
- Work on computer graphics, robot kinematics, and network implementation.

Depth and Flow Research – Computer Vision Course Project

March 2018 – June 2018

- Reproduced a few deep learning methods for depths and flow and evaluate their performances on a robotic setting dataset.

SoundScape Project – UW Adaptive Computing Machines and Emulator Lab

March 2018 – June 2018

- Designed a sound processing pipeline from scratch for modelling the acoustics of a train whistle sound.

uMessage – Data Structure Course Project

Jan 2018

- Implemented various data structures that would drive word suggestions, spelling corrections and autocompletions.

Random News Generator -- DubHacks

Oct 2017

- Implemented Markov Chain for a demo website that generates random news from large resources on a given topic

Optical Manipulation Project – UW Robotics and States Estimation Lab

Aug 2017 – March 2018

- Worked as a research assistant with Dr.Yu to implement a manipulation demo for PoseCNN, which enables the robot Baxter to classify and manipulate objects.
- Studied and had experience with camera structure, robot kinematics, Baxter SDK and ROS packages.

Music Editor – CodeDay at Seattle

Feb 2017

- Created a demo app that can record sound pieces and translate into sheet music display UI using DTFT, and it was awarded Overall 1st place out of 35 teams in Seattle

Leadership and Involvement:

Advanced Robotics at the University of Washington – Computer Vision Lead & Drone Lead

Sep 2016 – Feb 2018

- Managed a team of 10 engineering students to design computer vision for our robots in a robotic competition RoboMasters hosted by DJI, one of the world's current leader in drone manufacturing
- Used Python, ROS, C++ to perform system integration of controls, navigation and serial communication
- Implement the detection, shooting, refilling, landing and communication using DJI SDK on DJI M100 drone

Skills:

Languages: Java, C++, C, Python

Platform & Frameworks: ROS(Robot OS), Ubuntu, ARM microcontrollers, Arduino, Nvidia TX Series, STM32

Tools: Matlab, Git, Cmake, OpenCV, DJI_Guidance, Bash, Quartus, Latex, Tensorflow, Pytorch, MXNet

Research Interests: Computer Vision, Deep Learning, Robotics